

LI-KANG (TONY) WENG

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EDUCATION

The University of Texas at Dallas <i>Master of Science</i> in Computer Science	Dallas, USA 05/2021
Udacity <i>Nanodegree</i> in Self-Driving Car Engineer	Taipei, Taiwan 02/2018
• Projects Vehicle Tracking, Lane Line Detection, Traffic Sign Classification, Driving Behavior Cloning	
National Taiwan University (NTU) <i>Master of Science</i> in Bio-Industrial Mechatronics Engineering (BIME)	Taipei, Taiwan 09/2015
• Dissertation Sensor Fusion of Stereo Vision and Radar Systems for Vehicle Safety Application	
<i>Bachelor of Science</i> in BIME	06/2013
• Topic Quantitative Evaluation of the Floral Shape Variation in <i>Sinningia Speciosa</i> Domestication	

TECHNICAL SKILLS

Programming C++, Java, JavaScript, Python, Git, OpenCV, AWS
Robotic Components ROS, IMU, stereo camera, Nvidia JETSON TX2 Developer Board, Arduino

WORK EXPERIENCE

Software Development Engineer , Amazon Web Services, Inc.	Seattle, USA 06/2021 - present
• Contributed to EC2 Fleet API	
Software Development Engineer Intern , Amazon Web Services, Inc.	Dallas, USA 05/2020 - 07/2020
• Added CloudWatch Events to AWS EC2 Fleet to let customers be aware of the states of their fleets and take actions to fulfill their business demands	
Research Assistant , National Taiwan University	Taipei, Taiwan 03/2019 - 06/2019
• Built a tracked robot using ROS (Robot Operating System) for surveillance on a chicken farm and the information and the manipulation of the robot were visualized on GUI	
Software Engineer , HTC Corp.	Taipei, Taiwan 01/2017 - 07/2018
• Developed real-time visual-inertial SLAM algorithm using C++ on ARM platform	
• Designed GUI and tools for real-time data visualization and KPI measurement to eliminate tedious labor and quantitate tracking performance	

RELEVANT PROJECTS AND RESEARCH

Sensor Fusion Project , Biophotonics and Bioimaging Laboratory (BBLab), NTU	05/2013 - 09/2015
• Constructed sensor fusion based vehicle safety real-time system capable of obstacle detection, tracking and collision avoidance algorithms using stereo vision and millimeter-wave radar sensor	
• Overhauled algorithms and analyzed the performances to eliminate false detection of algorithm	
• Eliminated measurement error of depth information from 2.4% to 0.7% using fused information	
• Enhanced obstacle matched rate from 82.1% to 89.8% using fused information	
• Accelerated 2.8 times in the correspondence matching method using CUDA with OpenCV	
The 9th Utechzone Machine Vision Prize , Utechzone Inc.	02/2014 - 08/2014
• Awarded 2nd prize of the overall competition and developed fall detection algorithm including background removal, feature extraction, object tracking, and motion detection under complicated scenarios (light variation, overlap)	
Floral Shape Variation Study , BBLab, NTU	08/2012 - 02/2014
• Accelerated process speed and eliminated measurement error by developing a semi-automatic program with GUI using image processing methods for flower landmark acquisition	
• Analyzed shape variation of <i>Sinningia speciosa</i> from landmarks identified on 2D images	
Advanced Technology Project in Vehicle Safety: Intelligence and Human Factors , ARTC	05/2013 - 01/2014
• Refactored obstacle matching algorithm to optimize performance by speed-up around 30%	
• Eliminated measurement error by optimizing camera calibration on stereo vision	
• Analyzed path planning algorithms to provide more realistic solution	
• Designed GUI with concise information for user easy to understand environmental information	
The 8th Utechzone Machine Vision Prize , Utechzone Inc.	01/2013 - 08/2013
• Implemented feature of face recognition method and analyzed the performance with 78% successful rate	

SELECTED PUBLICATIONS

Ta-Te Lin, **Li-Kang Weng** and An-Chih Tsai. 2014. Object Tracking and Collision Avoidance Using Particle Filter and Vector Field Histogram Methods. Paper presented at ASABE. Paper No. 1906189, Montreal, Quebec City, Canada.